A Linear-Time Attention Kernel from Quantum Circuits: Recall Gains, Training-Cost Limits and Hardware Evidence

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***Abstract*—Quantum self-attention computes Transformer attention scores as overlaps of quantum states. We ask whether this is worth doing on quantum hardware, and what classical Transformers should borrow from it. We measure two scaling laws of the fidelity attention score: its mean decays as 2^{-n} and its gradient variance as approximately 4^{-n} in the number of qubits n (fitted rate 3.97, 95% interval 3.85–4.06); a second circuit design gives the same picture, and runs on IBM's `ibm_fez` processor reproduce the simulated scores ($r = 0.998$). Combining them with Bernoulli shot noise and parameter-shift gradients, we prove a lower bound on the shots per training step and a closed-form condition for any quantum advantage over exact classical simulation. The minimum training time inside any advantage regime scales as κ^2/R_s^2, where κ is the number of circuits per token pair and R_s the shot rate. With parameter-shift gradients this is at least 64 years per step on one sequence at 10^4 shots/s and 1.5 days at 10^6. A sweep over 144 hardware scenarios shows that training becomes practical only if three conditions hold together: a backpropagation-like gradient estimator, which no known construction provides for this circuit; coherent aggregation of gradients across token pairs; and at least 10^5 shots/s. On the classical side, we give an exact 3^n-dimensional feature map for the cosine-product kernel that underlies fidelity attention and evaluate the resulting linear-time layer, CPLA, on multi-query associative recall in two-layer Transformers. At an identical 81-dimensional state, CPLA retains 86% recall at 16 key-value pairs where standard linear attention falls to 12.5%, and it far exceeds Performer; it trails the second-order Taylor kernel of Based by 4–13 points at moderate load, and no fixed-state method survives 32 pairs. On character-level WikiText-2, by contrast, CPLA brings no gain over other linear layers, so its benefit is specific to recall. The study is released as a tested, resumable hybrid cloud–quantum pipeline whose automated checks tie every reported number to the committed data. Quantum attention thus contributes a multiplicative kernel useful for recall; a speedup would require gradient estimators and shot rates beyond those available today.**

***Index Terms*—Transformers, attention mechanism, linear attention, associative recall, quantum machine learning, quantum kernels, barren plateaus, computational cost**

I. Introduction

Attention is the most expensive part of a Transformer [1]. Its pairwise scores cost time quadratic in the sequence length N, which has motivated a large body of work on efficient attention: kernel views of attention [2], linear attention [3], random-feature approximations [4] and re-weighted linear kernels [5]. A recurring finding is that efficient models struggle with *associative recall*, retrieving a value seen earlier in the context given its key [6], [7].

Quantum self-attention has been proposed as another route [8]–[13]. Tokens are encoded as quantum states, and attention scores are computed from state overlaps on a quantum processor. Our earlier work [14] showed that, for angle-encoded states, the fidelity score equals a classical cosine-product (CP) kernel, and that any fidelity attention admits an exact linear-time classical evaluation. Two questions remain for anyone building Transformers.

1) Is training quantum attention on hardware ever worth its cost? Existing analyses estimate shot counts for evaluating quantum kernels [15], [16], but not the cost of *training* an attention layer, where every token pair needs gradient estimates.
2) What, if anything, should classical Transformers borrow? If the useful part of quantum attention is a classical kernel, does that kernel help on the recall tasks where efficient attention is known to fail?

We make five contributions.

- We measure two scaling laws for the quantum fidelity score: its mean decays as 2^{-n} and its gradient variance as $\approx 4^{-n}$ in the number of qubits n. Both hold for a second circuit design, and the score's behaviour is confirmed on IBM quantum hardware (Section IV).
- We prove a training-cost theorem. Accounting for Bernoulli shot noise, the number of circuits per gradient and the most quantum-favourable aggregation of gradients, it gives a lower bound on the shots per training step and a closed-form condition for any quantum advantage over exact classical simulation (Section V).
- We put numbers on the bound. With parameter-shift gradients, any regime in which quantum hardware is cheaper than exact classical simulation needs at least 1.5 days per training step on a single sequence at 10^6 shots/s and 64 years at 10^4; an idealized gradient estimator would shorten this to seconds at 10^6 shots/s, and we quantify

both cases over 144 hardware scenarios.

- We give an exact 3^n-dimensional feature map for the CP kernel, smaller than the 4^n of [14], and evaluate the resulting linear-time layer (CPLA) against Based [7], linear attention and Performer on multi-query associative recall, where it helps, and on character-level WikiText-2, where it does not (Section VI).
- We release the study as a tested, resumable hybrid cloud–quantum pipeline in which 36 automated checks tie every reported number to the committed data (Section VII).

II. Related Work

A. Efficient Attention and Recall

Tsai *et al.* [2] interpret attention as kernel smoothing, which lets the similarity function be studied apart from the projections around it. Linear attention [3] uses a positive feature map so that the sum over keys can be reassociated, giving cost linear in N; Performer [4] estimates softmax attention through positive random projections, and cosFormer [5] modulates scores with a cosine of token distance. Zoology [6] introduced multi-query associative recall (MQAR) and showed that efficient architectures lag attention on it. Based [7] pairs a small sliding window with linear attention whose features expand $\exp(q \cdot k)$ to second order, trading recurrent-state size for recall.

B. Quantum Self-Attention

QSANN [8] scores queries and keys from single Pauli-Z measurements; QKSAN [9] uses quantum kernel overlaps; QMSAN [10] uses swap tests between mixed states; Pecilli and Rosati [11] drop the softmax to obtain gate complexity linear in sequence length; and Xu *et al.* [12] build higher-order interactions from expectation read-outs. A recent survey [13] observes that scoring token pairs one at a time on a quantum device does not escape the N^2 cost of classical attention. Our prior work [14] reduced fidelity attention exactly to classical computation. None of these works analyses the end-to-end cost of training.

C. Cost and Trainability of Quantum Kernels

Fidelity kernels [17] can concentrate exponentially, so that distinguishing inputs needs exponentially many shots [16]. Slattery *et al.* [18] found that well-tuned fidelity kernels are approximated by classical kernels, and Bowles *et al.* [19] found no systematic advantage of quantum models in benchmarks. Miroszewski *et al.* [15] estimated the shots needed to *evaluate* quantum kernel matrices and compared them with classical simulation, concluding that quantum utility may appear above about 40 qubits. Bowles *et al.* [20] showed that circuits with commuting generators admit gradient estimation with far fewer circuits than the parameter-shift rule. Barren plateaus [21], particularly for global observables [22], make gradients vanish exponentially, and their provable absence often coincides with classical simulability [23]. Our analysis is specific to training an attention layer: it counts all N^2 token pairs, includes the cost of estimating gradients and compares against the linear-time classical reduction.

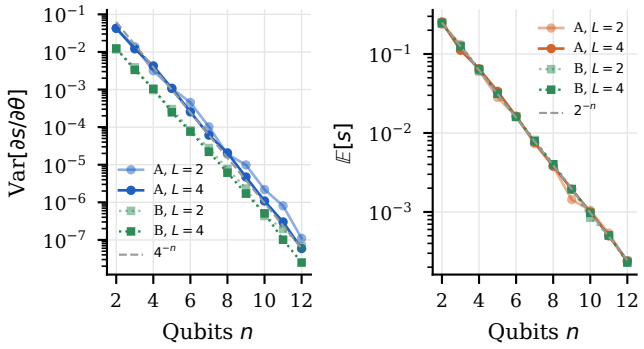


Fig. 1. Gradient variance (left) and mean score (right) of the fidelity attention score at uniformly random parameters for $L = 2$ and $L = 4$ ansatz layers: design A (Section III, 400 samples per point) and design B (R_X encoding, $R_Y R_Z$ layers, CZ chain; 1,600 samples per point). Dashed lines show 4^{-n} and 2^{-n}.

III. Background: Quantum Fidelity Attention

Each token embedding x is mapped to n angles $\phi(x) = \pi \tanh(Ax)$ and encoded as $S(\phi)\,|0\rangle^{\otimes n} = \bigotimes_k (\cos\frac{\phi_k}{2}\,|0\rangle + \sin\frac{\phi_k}{2}\,|1\rangle)$. Separate hardware-efficient ansatzes W_Q, W_K with L layers of R_Y rotations and CNOT rings give query and key states, and the score is the fidelity

$$s_{ij} = |\langle \psi_K(x_j) | \psi_Q(x_i) \rangle|^2 , \tag{1}$$

estimated on hardware with a compute–uncompute circuit whose all-zeros outcome probability equals s_{ij} [14]. Each score circuit has $P = 2n + 2nL = 2n(L+1)$ rotation angles, and gradients follow the parameter-shift rule $\partial s/\partial a = \frac{1}{2}[s(a + \frac{\pi}{2}) - s(a - \frac{\pi}{2})]$ [24], so a training step evaluates $\kappa = 1 + 2P$ circuits per token pair. We keep κ general below, since better gradient estimators can lower it [20].

Without an ansatz the score is the CP kernel $s_{ij} = \prod_k \cos^2\big((\phi_{ik} - \phi_{jk})/2\big)$, and with any ansatz s_{ij} can still be written as a dot product of 4^n-dimensional feature vectors [14]. We sharpen the feature dimension for the CP kernel.

Proposition 1 (Exact 3^n feature map)**.** *Let* $f(\phi) = \bigotimes_{k=1}^{n} \frac{1}{\sqrt{2}}\,[1, \cos\phi_k, \sin\phi_k] \in \mathbb{R}^{3^n}$. *Then* $\prod_k \cos^2\big((a_k - b_k)/2\big) = \langle f(a), f(b) \rangle$.

Proof. For one feature, $\cos^2\frac{a-b}{2} = \frac{1}{2}\big(1 + \cos(a-b)\big) = \frac{1}{2}\big(1 + \cos a \cos b + \sin a \sin b\big)$, the inner product of $\frac{1}{\sqrt{2}}[1, \cos a, \sin a]$ and $\frac{1}{\sqrt{2}}[1, \cos b, \sin b]$. Inner products of tensor products factorize. □

The map reduces the linear-time state from 4^n to 3^n per head (81 instead of 256 for $n = 4$), and allows separate query and key angle projections while preserving exact linear-time evaluation. We call the resulting layer cosine-product linear attention (CPLA).

IV. Two Scaling Laws of the Fidelity Score

We measured, for $n = 2$ to 12 and $L \in \{2, 4\}$, the mean score $p(n) = \mathbb{E}[s]$ and the gradient variance $V(n) =$

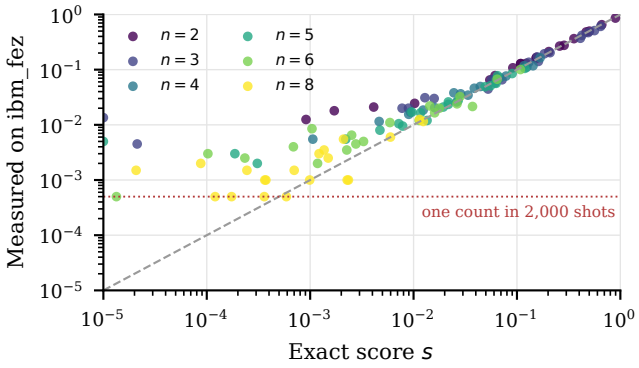


Fig. 2. Fidelity scores measured on IBM `ibm_fez` (2,000 shots per circuit) against their exact values, for 20 random circuits per register size (design A, $L = 2$). Scores below one count per 2,000 shots (dotted) cannot be resolved, which is the shot cost of Lemma 1 in practice.

$\mathrm{Var}[\partial s/\partial\theta_{Q,1}]$ over uniformly random inputs and parameters, using an exact statevector simulator validated against the Qiskit Estimator [25] to within 10^{-15}. Fitting log-linear models for $n \geq 4$ gives (95% bootstrap intervals in parentheses):

- $p(n) \propto 1.99^{-n}$ ($L = 2$) and 2.01^{-n} ($L = 4$), matching the Haar value $\mathbb{E}|\langle\psi|\varphi\rangle|^2 = 2^{-n}$;
- $V(n) \propto 3.53^{-n}$ (3.27–3.83) for $L = 2$ and 3.97^{-n} (3.85–4.06) for $L = 4$.

The deeper circuit reaches the 4^{-n} rate expected of a global observable on an approximate 2-design [21], [22].

A second circuit design. To check that these rates are not specific to one circuit, we repeated the measurement for a design B that changes every gate type: R_X angle encoding, layers of R_Y and R_Z rotations on every qubit (complex amplitudes, $2nL$ parameters per state) and a CZ chain in place of the CNOT ring, with 1,600 samples per point and the simulator checked against Qiskit to within 10^{-15}. The mean score again decays as 2.01^{-n} at both depths, and the gradient variance as 3.39^{-n} (3.31–3.58) for $L = 2$ and 3.70^{-n} (3.52–3.84) for $L = 4$, somewhat more slowly than for the first design (design A). Fig. 1 shows both designs. The cost analysis below uses both, and we state their extrapolation beyond 12 qubits as an assumption.

Hardware check. We ran the compute–uncompute circuit of design A ($L = 2$) on IBM's 156-qubit Heron processor `ibm_fez` for $n \in \{2, 3, 4, 5, 6, 8\}$, with 20 random input–parameter pairs per size and 2,000 shots per circuit (120 circuits; Fig. 2). Across all circuits the measured scores correlate with the exact values at $r = 0.998$ ($r \geq 0.95$ for every n), with a mean absolute error of 0.007, and the measured mean decays as 2.09^{-n} for $n \geq 4$, against 2.22^{-n} for the exact values of the same circuits. At $n = 8$ the binomial standard error of a single 2,000-shot estimate (0.0008) is already about a third of the mean score, the shot cost that Lemma 1 formalizes.

Assumption 1. For the register sizes considered, $p(n) = c_p\, r_p^n$ and $V(n) = a\, r_V^n$ with the fitted constants, and a typical per-pair gradient has magnitude $\sqrt{V(n)}$.

V. The Cost of Training Fidelity Attention

We compare one training step (forward and backward) on one sequence of N tokens. Let R_s be the hardware shot rate, F the classical floating-point rate, d the value width and $g = n(L+1)$ the gates per state.

Lemma 1 (Shot noise)**.** *A score estimated from S shots has variance $p(1-p)/S$, and the parameter-shift estimate $\hat{g} = \frac{1}{2}(\hat{s}_+ - \hat{s}_-)$ has variance $p(1-p)/(2S)$.*

Proof. Each shot is a Bernoulli trial with success probability $s \approx p$. The two shifted estimates are independent, so $\mathrm{Var}\,\hat{g} = \frac{1}{4} \cdot 2\,p(1-p)/S$. □

Theorem 1 (Quantum shots per step)**.** *Under Assumption 1, resolving the gradient of a shared parameter at signal-to-noise ratio at least one requires, per training step,*

$$Q \;\geq\; \kappa\, \max\Big(N^2,\; \frac{p(1-p)}{2V}\Big) \tag{2}$$

shots, even if per-pair gradients add perfectly coherently.

Proof. Each of the N^2 pairs needs at least one shot on each of its κ circuits, giving the first term. The aggregated gradient is $G = \sum_{ij} w_{ij}\hat{g}_{ij}$, with noise variance $\sum w_{ij}^2\, p(1-p)/(2S)$ by Lemma 1. Its signal satisfies $|\mathbb{E}G|^2 \leq (\sum |w_{ij}|)^2 V \leq N^2 \sum w_{ij}^2 V$ by Cauchy–Schwarz, with equality for perfectly coherent pairs. Signal-to-noise at least one therefore needs $S \geq p(1-p)/(2N^2V)$ shots per circuit, and the total is $N^2\kappa S$. □

If per-pair gradients are incoherent, the signal grows only as N and the second term in (2) is multiplied by N^2.

Theorem 2 (Exact classical cost)**.** *Exact forward and backward evaluation of fidelity attention takes at most*

$$T_c = \frac{6}{F}\min\big(2Ng2^n + 2N^2 2^n,\; 2Ng2^n + 4N\,4^n d\big) \tag{3}$$

seconds, with memory 2^{n+5} bytes for the first (pairwise) algorithm and $8 \cdot 4^n d$ bytes for the second (linear-time) algorithm.

Proof. Preparing a state costs g gate applications on 2^n amplitudes. The pairwise algorithm computes N^2 overlaps of length 2^n, streaming states so that two are held at once. The linear-time algorithm uses the explicit 4^n features of [14] with prefix sums. Reverse-mode differentiation costs at most twice the forward pass, giving the factor $6 = 2 \times 3$ for multiply-adds over forward and backward. □

Corollary 1 (Necessary condition for advantage)**.** *If quantum hardware is faster than the pairwise classical algorithm, then $2^n(1 + g/N) > \kappa F/(12R_s)$.*

Proof. By Theorem 1, $T_q \geq \kappa N^2/R_s$; by Theorem 2, $T_c \leq 12(N^2 + Ng)2^n/F$. Require $T_q < T_c$. □

Corollary 2 (Minimum time per step)**.** *If $p = 2^{-n}$ and $V = a\,4^{-n}$, then whenever quantum hardware beats exact classical simulation with sufficient classical memory,*

$$T_q \;\geq\; \frac{\kappa^2 F}{24\, a\, R_s^2} \cdot \frac{N}{N+g}. \tag{4}$$

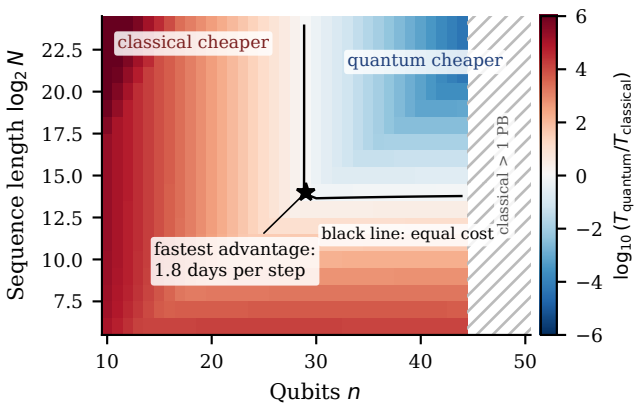


Fig. 3. Ratio of quantum to exact classical training time per step for $L = 4$, $R_s = 10^6$ shots/s, $F = 10^{13}$ flop/s and 1 PB classical memory. Quantum is cheaper only in the upper right, and the fastest such configuration (star) still needs 1.8 days per step on one sequence.

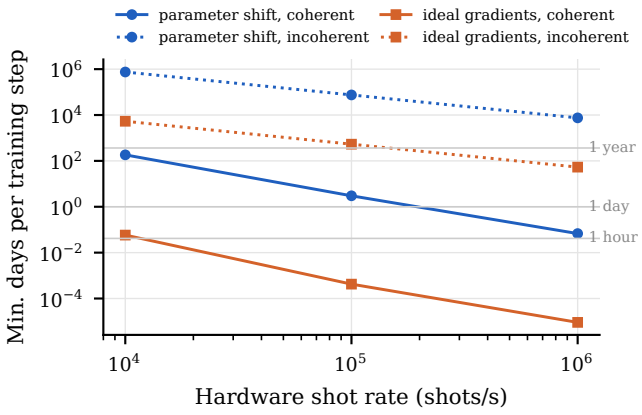


Fig. 4. Minimum time per training step over all configurations where quantum hardware beats exact classical simulation, versus hardware shot rate, for parameter-shift ($\kappa = 1 + 2P$) and idealized ($\kappa = 3$) gradient estimators with coherent or incoherent aggregation (sweep over L, classical flop rate 10^{13}–10^{17} and memory 10^{12}–10^{15} bytes).

Proof. By Theorem 1, $T_q \geq \kappa\, p(1-p)/(2VR_s) \approx \kappa\, 2^n/(2aR_s)$. Substituting the lower bound on 2^n from Corollary 1 gives (4). □

The bound grows as κ^2/R_s^2: both the circuit count and the shot rate enter quadratically, once through the per-pair floor and once through the gradient-shot requirement. With the measured constants for $L = 4$ ($a = 1.07$), $F = 10^{13}$ (one workstation) and parameter-shift gradients ($\kappa = 1 + 2P$), the smallest qualifying register and the resulting bound are $n = 36$ and $T_q \geq 64$ years per step at $R_s = 10^4$ shots/s; $n = 33$ and 198 days at 10^5; and $n = 29$ and 1.5 days at 10^6. A hypothetical estimator returning all gradients from two extra circuits ($\kappa = 3$) would lower the bound to 9.8 hours, 6 minutes and 3.5 seconds, at $n = 28$, 25 and 22. Known constructions of such estimators require commuting generators [20], which the hardware-efficient ansatz used here does not have.

Numerical sweep. To check the result without the idealized rates, we evaluated (2) and (3) with the fitted constants for every $n \leq 60$ and $N = 2^6$ to 2^{24}, across 144 scenarios: $L \in \{2, 4\}$, $R_s \in \{10^4, 10^5, 10^6\}$, $F \in \{10^{13}, 10^{15}, 10^{17}\}$, memory $\in \{1\,\text{TB}, 1\,\text{PB}\}$, coherent or incoherent gradients, and $\kappa \in \{1 + 2P, 3\}$. Fig. 3 maps the ratio for one scenario and Fig. 4 summarizes all of them. A quantum advantage region always exists at large n and N. With parameter-shift gradients, the fastest training step inside it takes at least 0.5 years at 10^4 shots/s, 3.0 days at 10^5 and 1.6 hours at 10^6 when gradients aggregate coherently, and 2,060, 206 and 20.6 years when they do not; a modest run of 10^4 steps is out of reach in every such scenario. With the idealized estimator and coherent gradients, the minima fall to 1.4 hours, 37 seconds and 0.8 seconds, so a 10^4-step run would take 1.6 years, 4.2 days or 2.2 hours; with incoherent gradients they remain 14.7 years, 1.5 years and 54 days per step. Training becomes practical only when a better gradient estimator, coherent gradients and at least 10^5 shots/s are all available. Repeating the sweep with the constants fitted for design B changes each of these twelve minima by at most a factor of 3.4 and leaves the conclusion unchanged.

VI. What Transformers Can Borrow: Associative Recall

If quantum attention is useful, the cost analysis points to its kernel. We therefore test the classical CP kernel where efficient attention is known to be weak: multi-query associative recall (MQAR) [6].

A. Setup

Each sequence lists K key–value pairs followed by $\min(K, 16)$ queries, each a previously seen key; the model must output the paired value. Keys and values come from disjoint vocabularies of 64 tokens each, so chance accuracy is $1/64$. Following Based [7], every model is a two-layer, four-head, pre-norm causal Transformer ($d = 64$) whose first layer is sliding-window softmax attention over the last 4 tokens and whose second layer is the attention variant under test. A control with sliding windows in both layers measures how much recall the global layer provides. The second layer is one of:

- softmax: full quadratic attention, the reference;
- CPLA: linear-time CP kernel with separate 4-angle query and key projections per head, state $3^4 = 81$ (Proposition 1);
- CP softmax: the same kernel inside a softmax, quadratic;
- Based [7]: second-order Taylor features of $\exp(q \cdot k/\sqrt{d'})$, state $1 + d' + d'^2 = 73$ ($d' = 8$) or 273 ($d' = 16$);
- linear elu+1 [3] with state 81, matched to CPLA;
- Performer [4] with 64 positive random features.

All models train for 2,000 steps of AdamW (batch 64, learning rate 10^{-3} with one-cycle schedule) on freshly sampled sequences and are evaluated on 2,048 held-out sequences, over three seeds for $K \in \{8, 16, 32\}$. Causality of every variant and exactness of the linear-time forms against their pairwise counterparts were verified numerically.

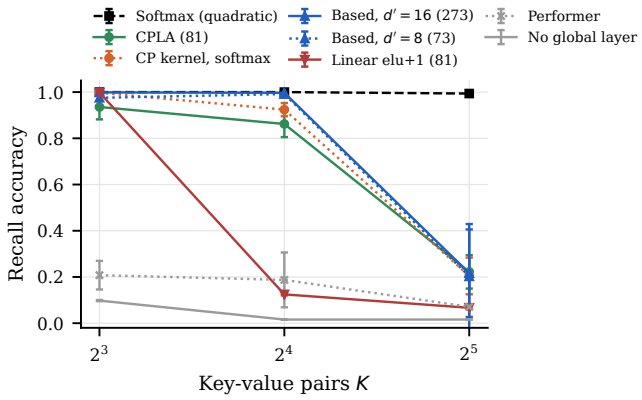


Fig. 5. Associative recall accuracy versus number of key–value pairs (mean ± std., 3 seeds). Numbers in parentheses are recurrent-state dimensions per head. At the same state size, CPLA (green) retains recall at $K = 16$ where linear elu+1 attention (red) collapses.

TABLE I
MULTI-QUERY ASSOCIATIVE RECALL ACCURACY (MEAN ± STD., 3 SEEDS)

Second layer	State	$K = 8$	$K = 16$	$K = 32$
Softmax (quadratic)	$O(N)$	1.000±.000	1.000±.000	.994±.002
CP softmax (quadratic)	$O(N)$	.997±.001	.924±.029	.204±.079
Based, d'=16	273	.999±.001	.997±.002	.216±.189
Based, d'=8	73	.975±.032	.991±.004	.203±.226
CPLA (ours)	81	.936±.054	.862±.057	.222±.072
Linear elu+1	81	.998±.001	.125±.004	.067±.002
Performer	64	.208±.062	.187±.119	.071±.002
No global layer	–	.098±.002	.016±.001	.016±.000

B. Results

The control without a global layer stays near chance (Table I, Fig. 5), so the second layer does the recall.

Kernel shape matters at fixed state size. CPLA and linear elu+1 attention carry the same 81-dimensional state per head, yet at $K = 16$ CPLA recalls 86.2% of values and elu+1 only 12.5%, a paired difference of 0.74 ± 0.06 over seeds. CPLA also exceeds Performer by 0.68 at $K = 16$. The CP kernel multiplies agreement across angle features, so a query matches a key only if all features align, whereas elu+1 features combine additively. Based's second-order Taylor kernel shares this multiplicative character, and both retain recall where additive features fail.

Based remains stronger. At moderate load Based outperforms CPLA by 4.0 points at $K = 8$ and 12.9 at $K = 16$ with a smaller 73-dimensional state; at $K = 32$ the two are indistinguishable. CPLA is also within 6.2 points of its own softmax version, so the linear-time form costs little relative to the kernel itself.

No fixed state survives high load. At $K = 32$ every linear-time variant falls to 7–22% while softmax attention keeps 99.4%, consistent with the recall–state-size trade-off of [6], [7]. At $K = 64$ even softmax attention reached only 40.7% within the 2,000-step budget, so we do not report that setting.

TABLE II
CHARACTER-LEVEL WIKITEXT-2: TEST BITS PER CHARACTER (MEAN ± STD., 2 SEEDS; LOWER IS BETTER)

Global layer	All characters	Copy positions
None (windows only)	1.899±.004	1.236±.006
Softmax (quadratic)	1.906±.001	1.241±.009
Performer	1.981±.002	1.343±.005
Linear elu+1 (81)	1.986±.007	1.355±.004
Based, d'=8 (73)	2.016±.002	1.379±.000
CPLA (81, ours)	2.081±.008	1.455±.018

C. Real Text: Character-Level WikiText-2

To test whether the recall advantage carries over to natural text, we trained four-layer, four-head models ($d = 128$, about 0.55M parameters) on character-level WikiText-2 with a 256-character context for 2,500 steps of AdamW (batch 32, peak learning rate 2×10^{-3}, one-cycle schedule), over two seeds. Layers alternate sliding-window softmax attention (window 64) with the global layer under test, and a control uses sliding windows in all four layers. Table II reports test bits per character (BPC) over the whole test split and on *copy positions*: the 16,753 characters (1.3%) that can be predicted by copying from an earlier occurrence of the same 8-character context at least 64 positions back.

No global layer helps at this scale, and CPLA is the weakest. The window-only control matches softmax attention, because four stacked 64-character windows already span the whole context, so this benchmark does not reward global retrieval. Among the efficient layers, CPLA trails linear elu+1 attention by 0.095 BPC overall and 0.100 on copy positions, and Based by 0.065 and 0.076. The advantage of CPLA on associative recall therefore does not transfer to character-level language modelling at this size, consistent with the finding of [14] that the CP kernel suits matching-type structure rather than graded similarity. Larger models with longer contexts, where global retrieval matters, remain to be tested.

VII. REPRODUCIBLE HYBRID WORKFLOW

Results that combine simulation, classical training and quantum hardware are easy to get subtly wrong and hard to rerun. We therefore built the study as a small, tested pipeline; this section describes how every number in the paper was produced and checked.

Claims as tests. A suite of 36 automated tests runs on GitHub Actions at every change and fails if the code or data drift from the paper. It checks the mathematics (Proposition 1 to 10^{-12}, and each linear-time form against its pairwise counterpart), a property no model may violate (causality of all seven attention variants), invariants of the cost model (κ and the lower bound of Theorem 1), and every reported number: the fitted rates of both circuit designs, twelve regenerated scenarios of the sensitivity sweep, the recall, WikiText-2 and hardware results, and the exact score of each hardware circuit, regenerated from its seed.

Deterministic, resumable experiments. All runs are seeded. Experiment drivers append one row per finished run and skip

TABLE III
COMPUTE–UNCOMPUTE CIRCUITS COMPILED FOR IBM_FEZ (DESIGN A, $L = 2$; MEAN OVER 20 CIRCUITS)

Qubits n	2	3	4	5	6	8
Depth	13	43	46	59	123	115
Two-qubit (CZ) gates	2	12	16	22	48	40

completed rows on restart; the recall grid also takes lock files, so parallel workers can share it without duplicating work, and the language-model runs save checkpoints. The cloud container restarted twice during this study; the drivers resumed without losing results, and a rerun model reproduced its earlier test BPC to within 10^{-7}. The recall, language-modelling and design-B experiments took 6.3 hours on a two-vCPU container.

Hybrid cloud–quantum execution. Hardware runs pass through three stages. Circuits are generated from a seed together with their exact scores; a dry run on a noiseless local sampler checks the pipeline end to end without spending hardware time; the same circuits are then compiled to the device's native gates with Qiskit's preset pass manager (optimization level 3) and submitted through Qiskit Runtime. Table III gives the compiled size. The cloud environment's network policy does not allow traffic to the quantum service, so the job was submitted from a local workstation, with the API key held in the local Qiskit account store and kept out of the repository and scripts. The 120-circuit job (240,000 shots) returned results 82 seconds after submission on IBM's free Open plan.

VIII. DISCUSSION AND LIMITATIONS

For AI practitioners. The useful part of quantum fidelity attention is its kernel. Its CP form is exactly computable in linear time with 3^n features per head. It helps on recall-style tasks but not, in our tests, on general character-level text, so its usefulness should be judged task by task against efficient-attention baselines at matched state size.

For quantum machine learning. The cost theorem isolates four levers: the circuits per pair κ, the per-pair floor $N^2\kappa$, the gradient-shot requirement $p/(2V)$, and the shot rate. Practical training would need progress on several at once: gradient estimators with backpropagation-like scaling for entangling ansatzes [20]; gradient variance decaying more slowly than the score mean squared, which in practice means local rather than global observables [22]; non-pairwise readouts that avoid N^2 circuits [11]; and shot rates of 10^5 per second or more.

Limitations. The scaling laws are measured in simulation up to 12 qubits, checked on hardware up to 8 qubits for one design, and extrapolated (Assumption 1). The cost model counts shots and floating-point operations and ignores error correction, compilation, communication and classical-quantum latency, all of which would add to the quantum side. The recall and language-modelling experiments use small models (two and four layers) trained on a CPU budget. The recall gap to softmax attention at $K = 32$ and the failure of all models at $K = 64$ reflect a fixed 2,000-step budget; longer training or larger states may change the ranking.

IX. CONCLUSION

With current gradient methods and hardware, quantum fidelity attention is useful as a kernel and does not offer a training speedup. Its score mean and gradient variance decay as 2^{-n} and 4^{-n} in two circuit designs and on IBM hardware, and with these laws we proved that any regime where hardware beats exact classical simulation demands a training time proportional to κ^2/R_s^2: from 1.5 days per step at 10^6 shots/s to 64 years at 10^4 with parameter-shift gradients. Only a backpropagation-like gradient estimator combined with high shot rates would bring training within reach. What Transformers can borrow is the cosine-product kernel itself: computed exactly in linear time with 3^n features, it retains associative recall at a fixed state size where additive linear attention collapses, although Based's Taylor kernel remains stronger; on character-level WikiText-2 it brings no gain, so its benefit is specific to recall. Because every result is regenerated and checked automatically, these conclusions can be rerun as hardware and gradient methods improve. Future work should combine CP and Taylor features, test the kernel in larger language models with longer contexts, and look for efficient gradient estimators and for local, non-pairwise readouts, since our bounds do not rule these out.

CODE AND DATA AVAILABILITY

Code, raw results and figure scripts for the scaling measurements, cost analysis and recall experiments are available at https://github.com/Chandra953/Quantum-Attention-Cost. The underlying quantum layer and its validation are in [14].

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